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CYBER DEFENSE OPERATIONS

Network Security

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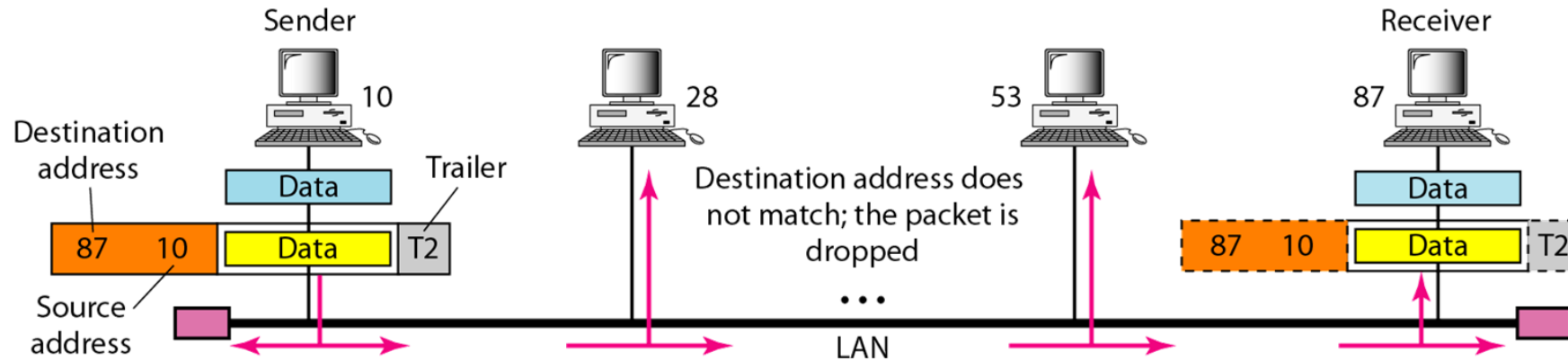
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REVIEW

NETWORK ADDRESSING

- Any communication that involves two parties needs two addresses:
 - Source address
 - Destination address
- Each device (computer or router) has a pair of addresses (**logical and physical**) for each connection

PHYSICAL ADDRESS (MAC ADDRESS)



- Most local-area networks use a **48-bit** (6-byte) physical address written as 12 hexadecimal digits
- Every byte (2 hexadecimal digits) is separated by a colon, as shown below

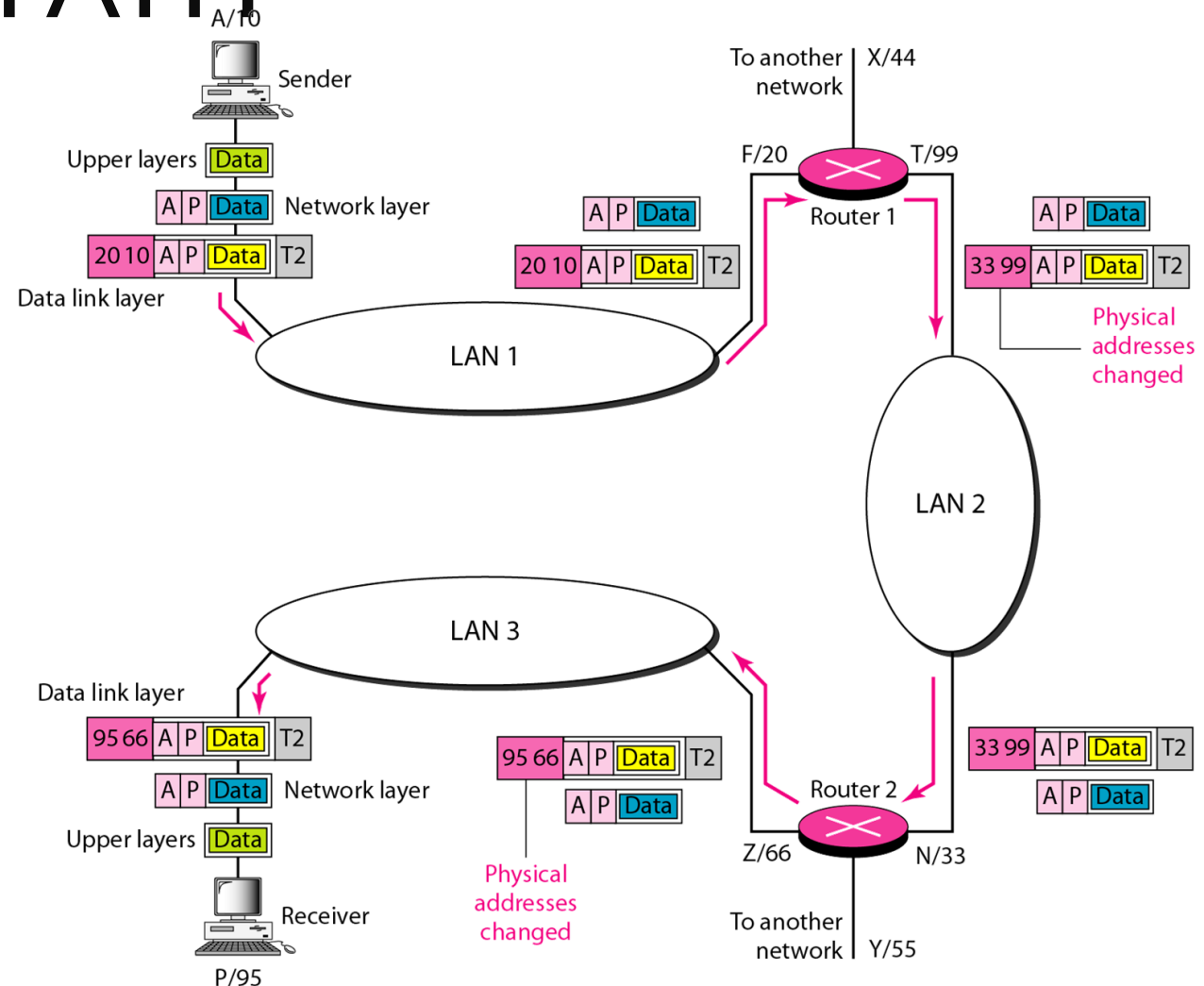
07:01:02:01:2C:4B

LOGICAL ADDRESS (IP ADDRESS)

- A **logical IP address** is a unique identifier assigned to devices on a network, enabling them to communicate with each other.
- **IPv4** (Internet Protocol Version 4)
 - 32-bit address divided into four octets
 - Example: **192.168.10.10**
- **IPv6** (Internet Protocol Version 6):
 - 128-bit address divided into eight groups of four hexadecimal digits
 - Example: **2001:0db8:85a3:0000:0000:8a2e:0370:7334**

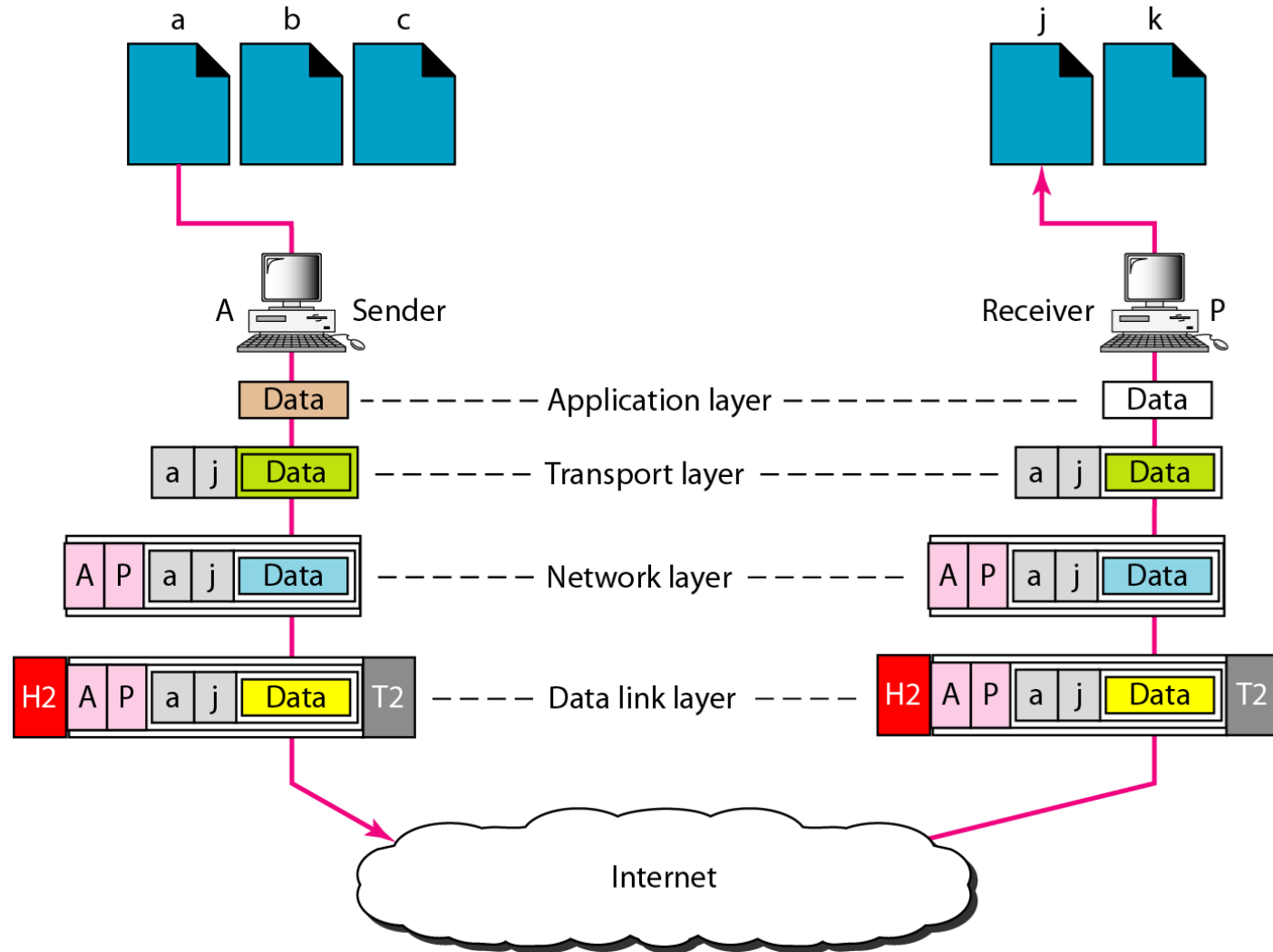
CHANGE OF ADDRESSES ALONG THE TRAVEL PATH

The physical addresses will change from hop to hop but the logical addresses usually remain the same.



PORT ADDRESS

- A **port number** (16-bit) is a numerical identifier used in networking to distinguish different services or applications on a single device.
- Port numbers **range** from 0 to 65535.
- **Well-Known Ports** (0-1023): Reserved for common services and protocols (e.g., HTTP uses port 80, HTTPS uses port 443, FTP uses port 21, DNS uses port 53).



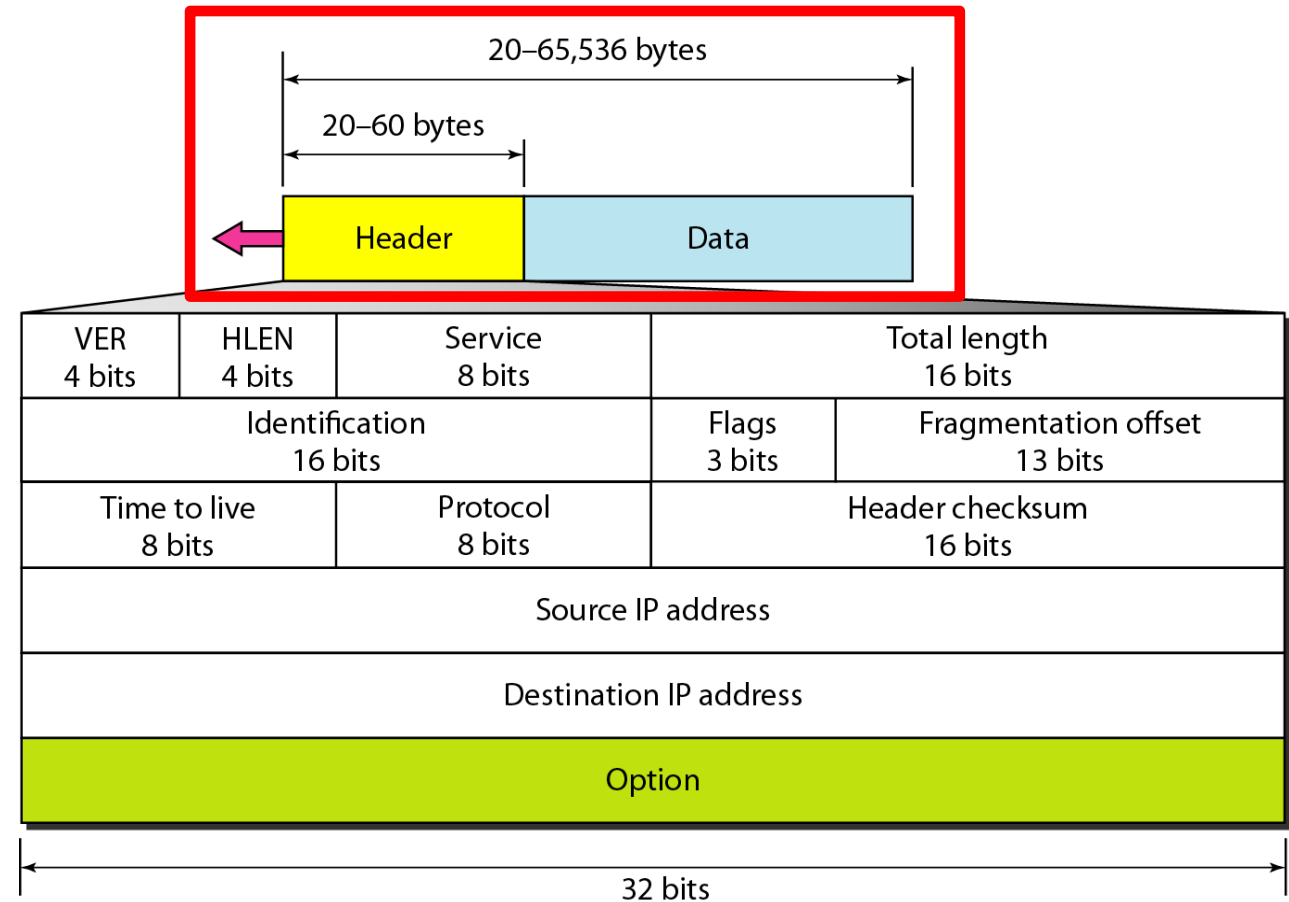
IP DATAGRAM (PACKET)

IP datagram is a variable-length packet consisting of two parts:

- Header
- Data

IP Header: 20 to 60 bytes in length

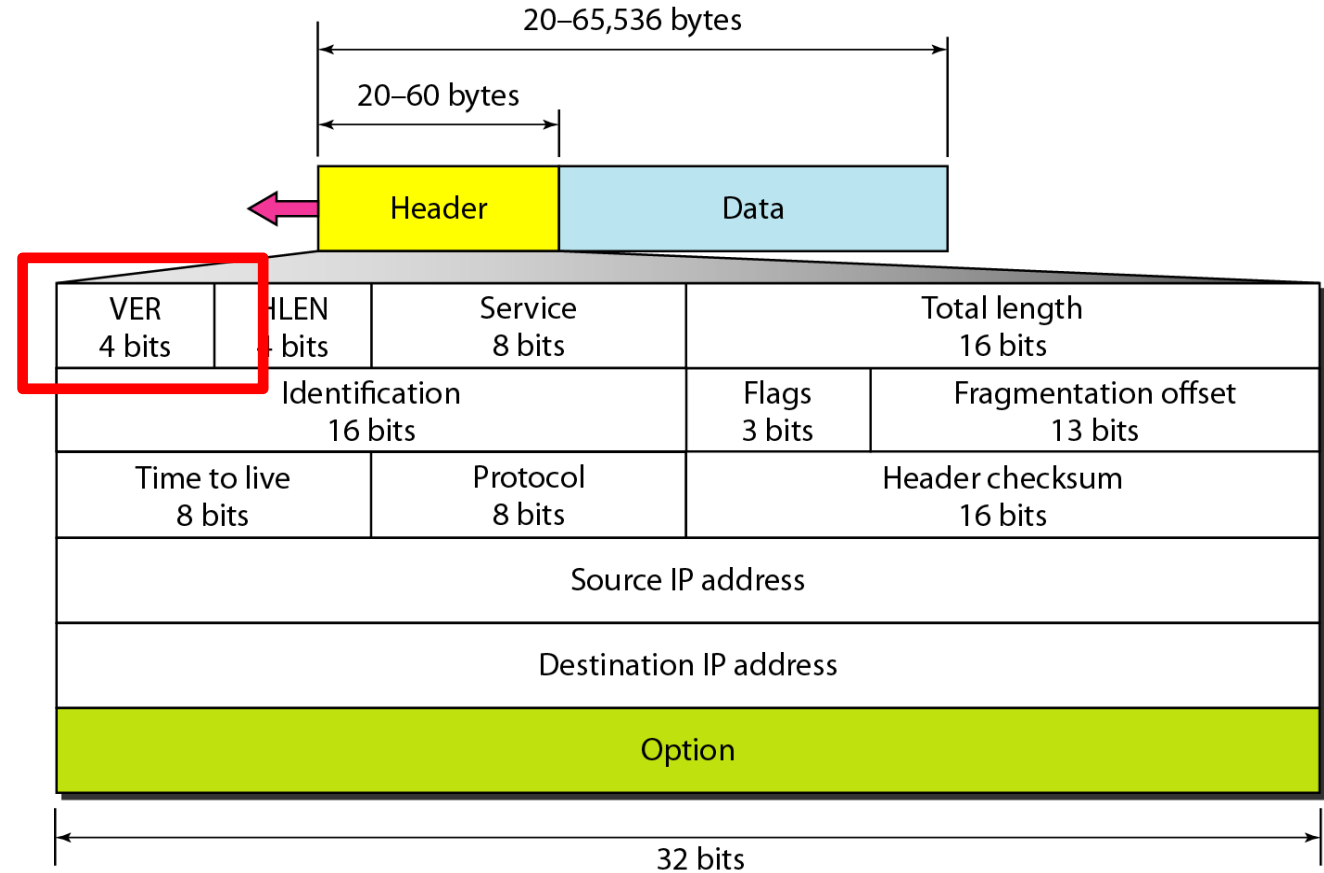
- Contains information essential to routing and delivery



VERSION, HEADER LENGTH AND TOTAL LENGTH FIELD

Version (4 bit)

- IPv4: 4, IPv6: 6



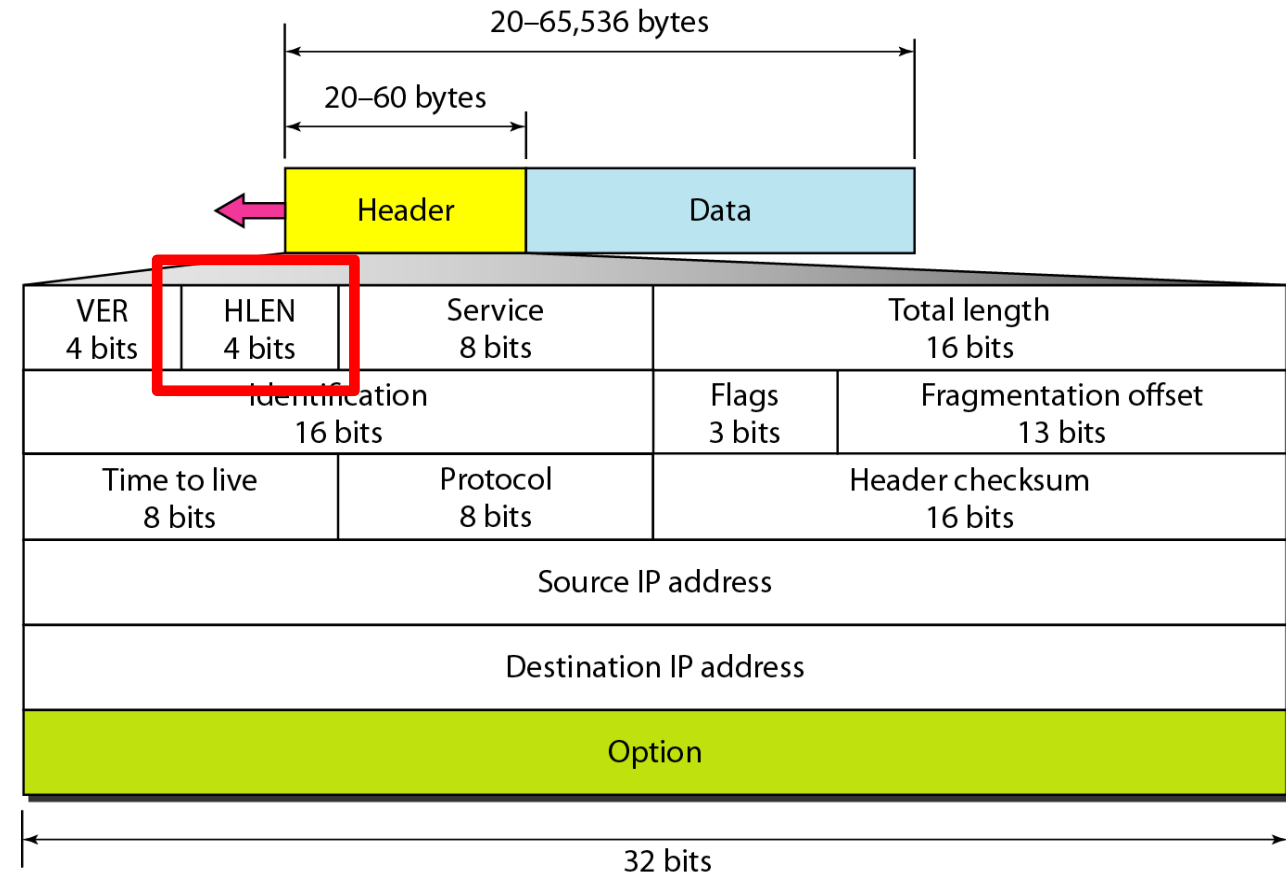
VERSION, HEADER LENGTH AND TOTAL LENGTH FIELD

Version (4 bit)

- IPv4: 4, IPv6: 6

Header length (4 bits)

- Indicates how long the header is, in 32 bit “words”. The minimum value is “5” which would be 160 bits, or 20 bytes.



VERSION, HEADER LENGTH AND TOTAL LENGTH FIELD

Version (4 bit)

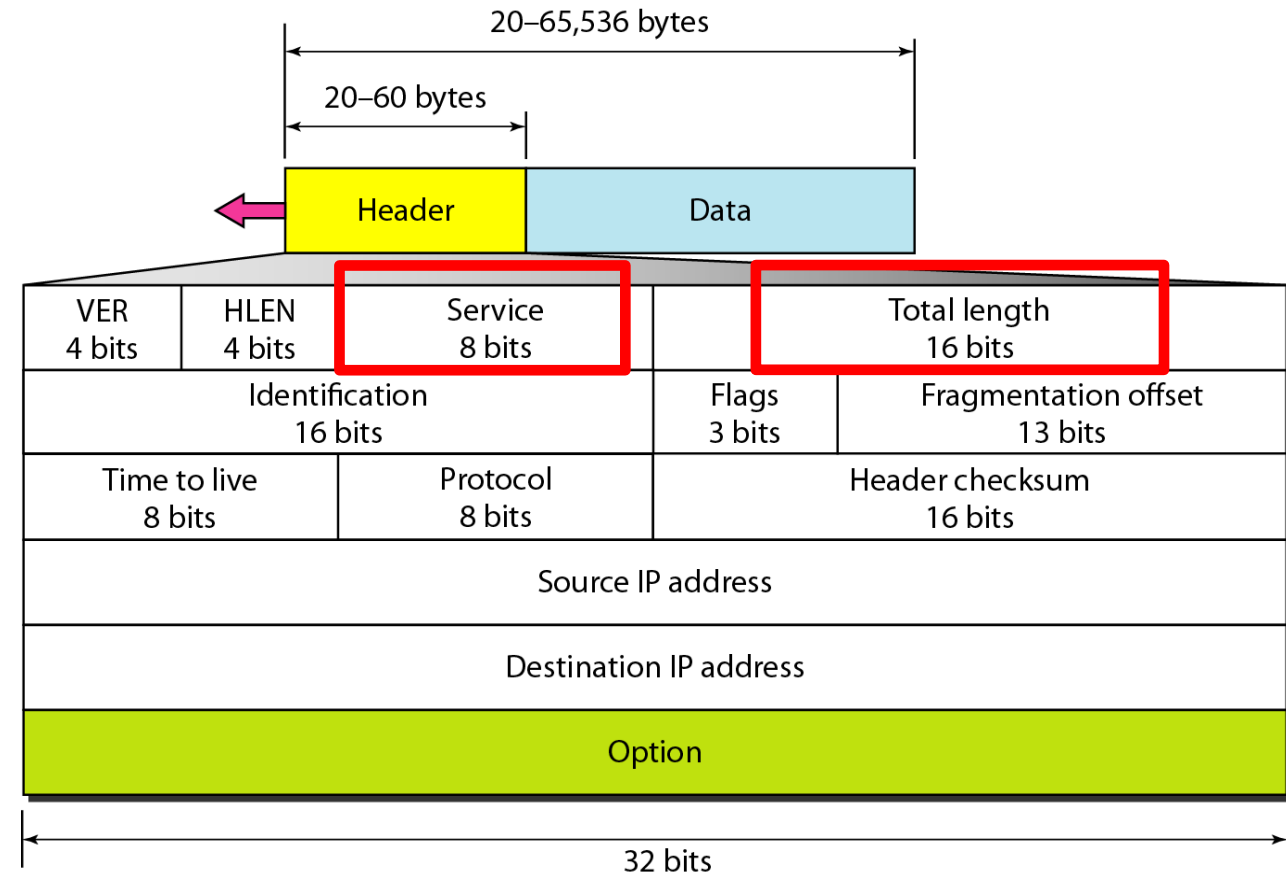
- IPv4: 4, IPv6: 6

Header length (4 bits)

- Indicates how long the header is, in 32 bit “words”. The minimum value is “5” which would be 160 bits, or 20 bytes.

Total Length field (16 bits)

- Contains the total length of the packet, including the packet header, in bytes. The minimum length is 20 (20 bytes of header plus 0 bytes of data), and the maximum is 65,535 bytes.

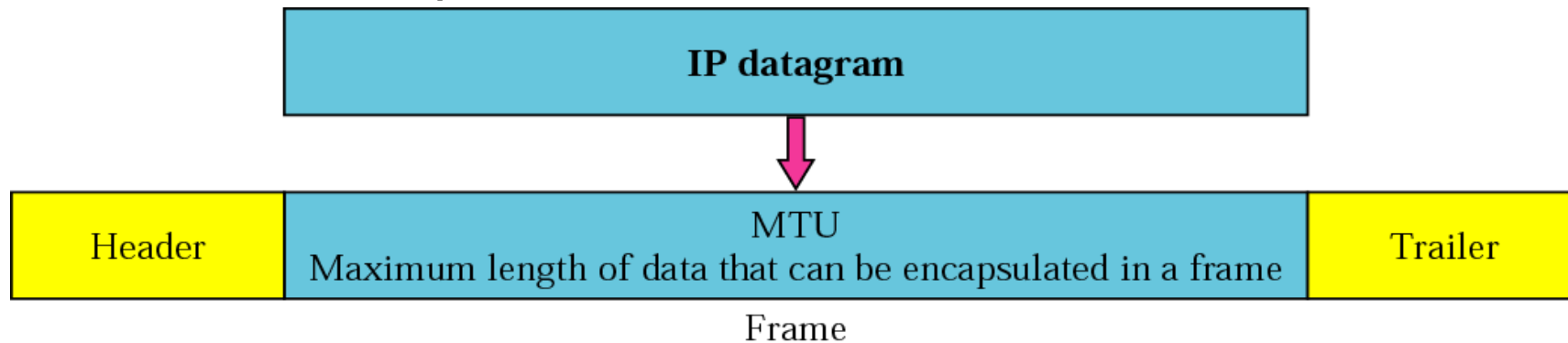


FRAGMENTATION

Format and size of a Layer-2 frame depends on the protocol used by the physical network - **Maximum Transmission Unit (MTU)**

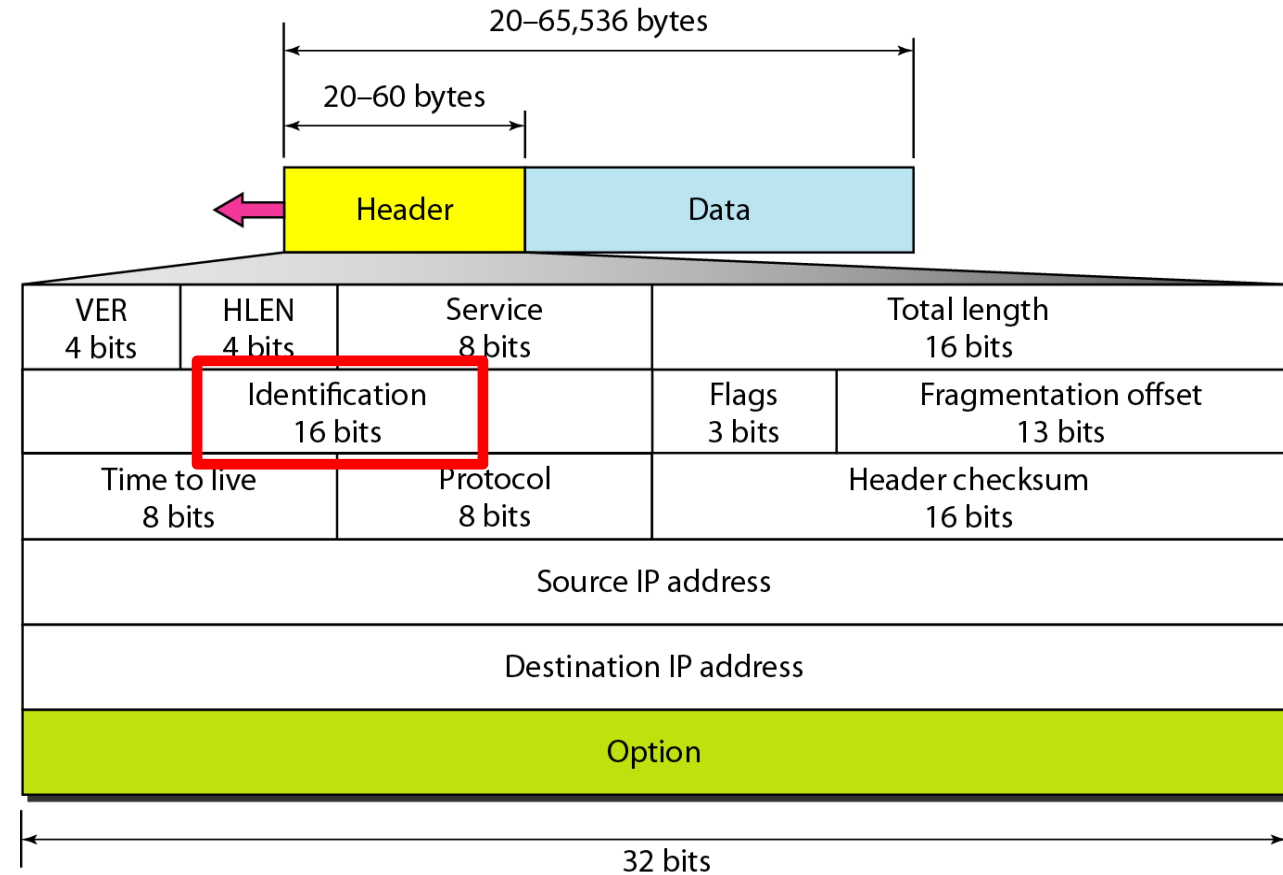
- Example: max payload size of Ethernet is 1500 bytes

When an IP packet **exceeds** the Maximum Transmission Unit (MTU) of the network, it is fragmented into smaller pieces.



IDENTIFICATION NUMBER

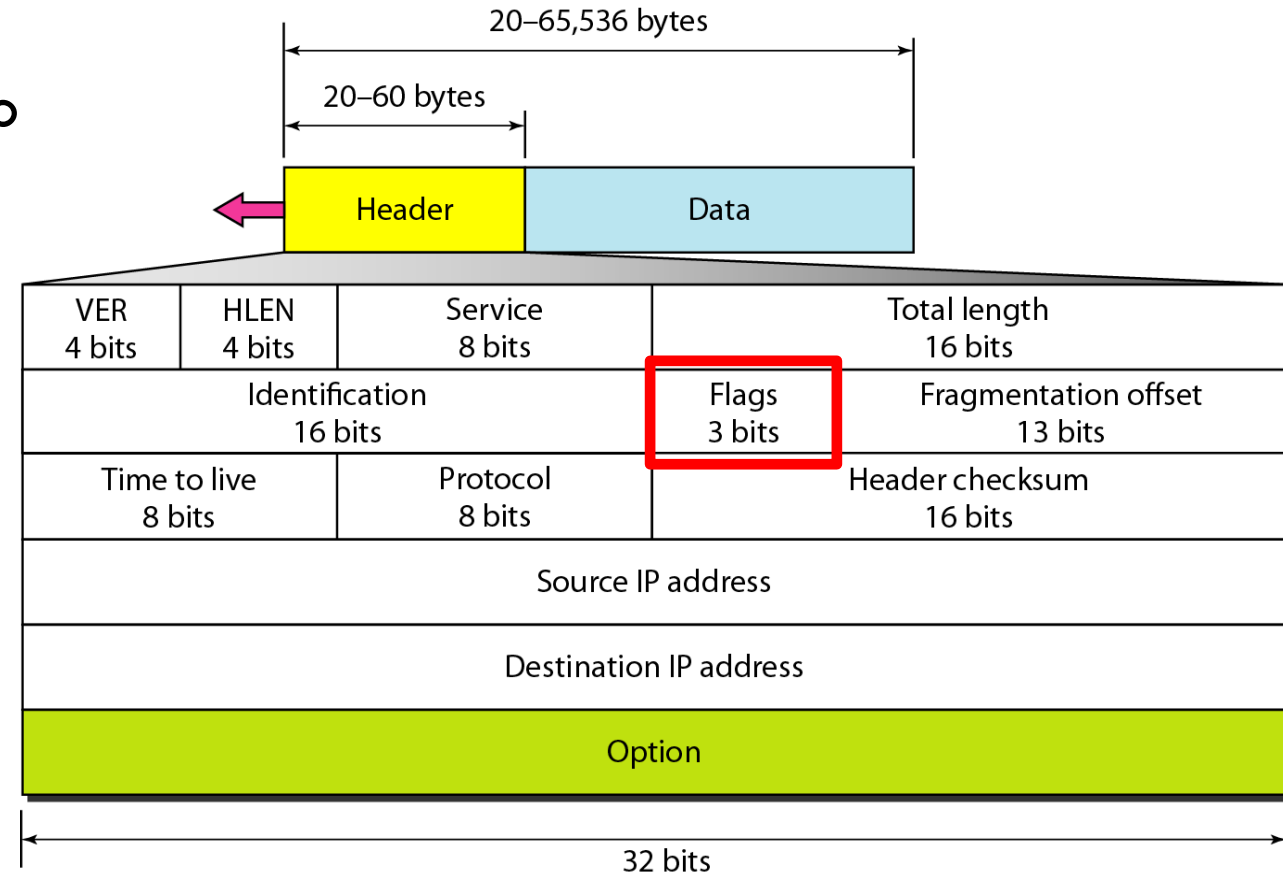
Each fragment is assigned the **same identification number** so that the receiving device knows which fragments belong together and can reassemble them into the original packet.



FLAGS AND FRAGMENTATION OFFSET

Flags: A 3-bit field follows and is used to control or identify fragments

- bit 0: Reserved; must be zero.
- bit 1: Don't Fragment (DF)
- bit 2: More Fragments (MF)

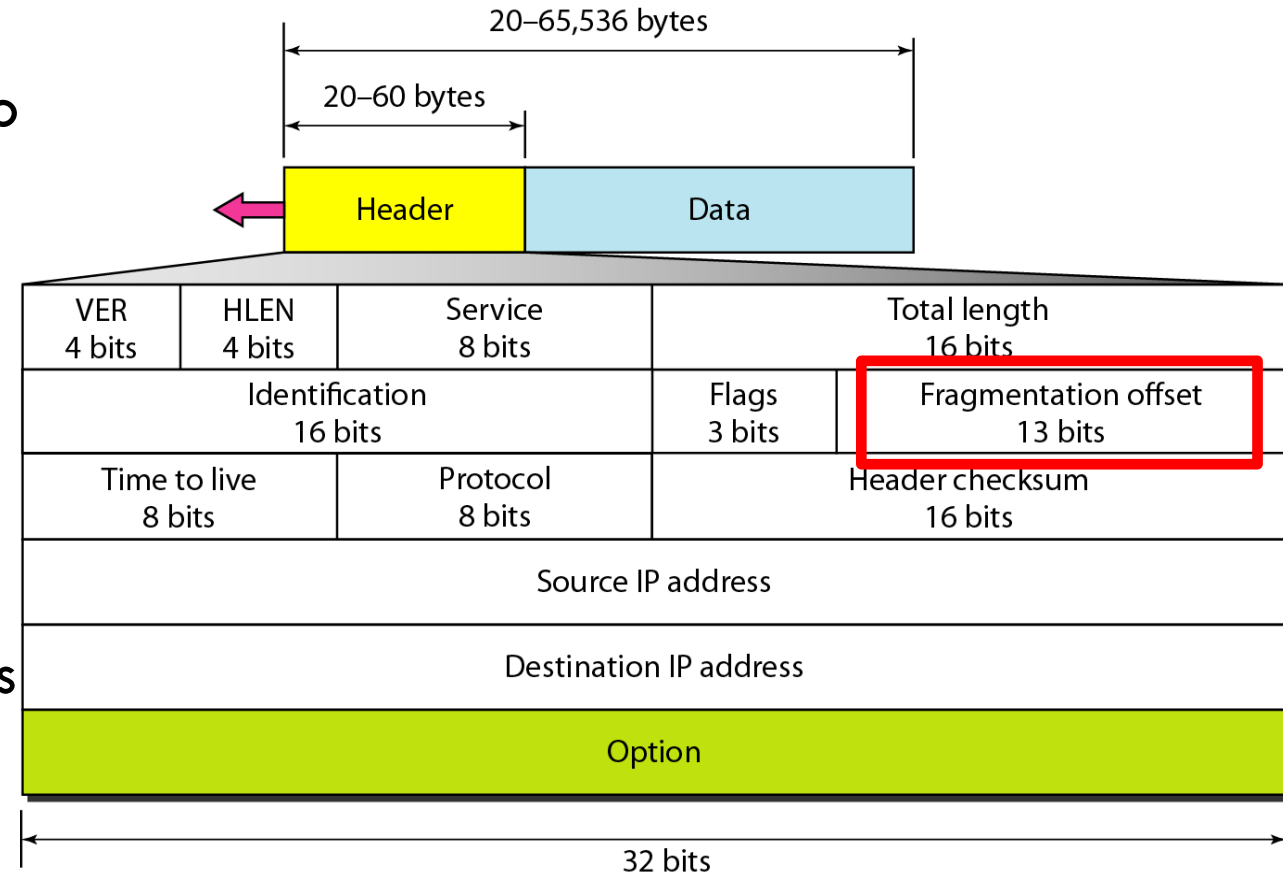


FLAGS AND FRAGMENTATION OFFSET

Flags: A 3-bit field follows and is used to control or identify fragments

- bit 0: Reserved; must be zero.
- bit 1: Don't Fragment (DF)
- bit 2: More Fragments (MF)

Fragment Offset: A 13-bit field indicates the offset of a particular fragment relative to the beginning of the original unfragmented IP datagram



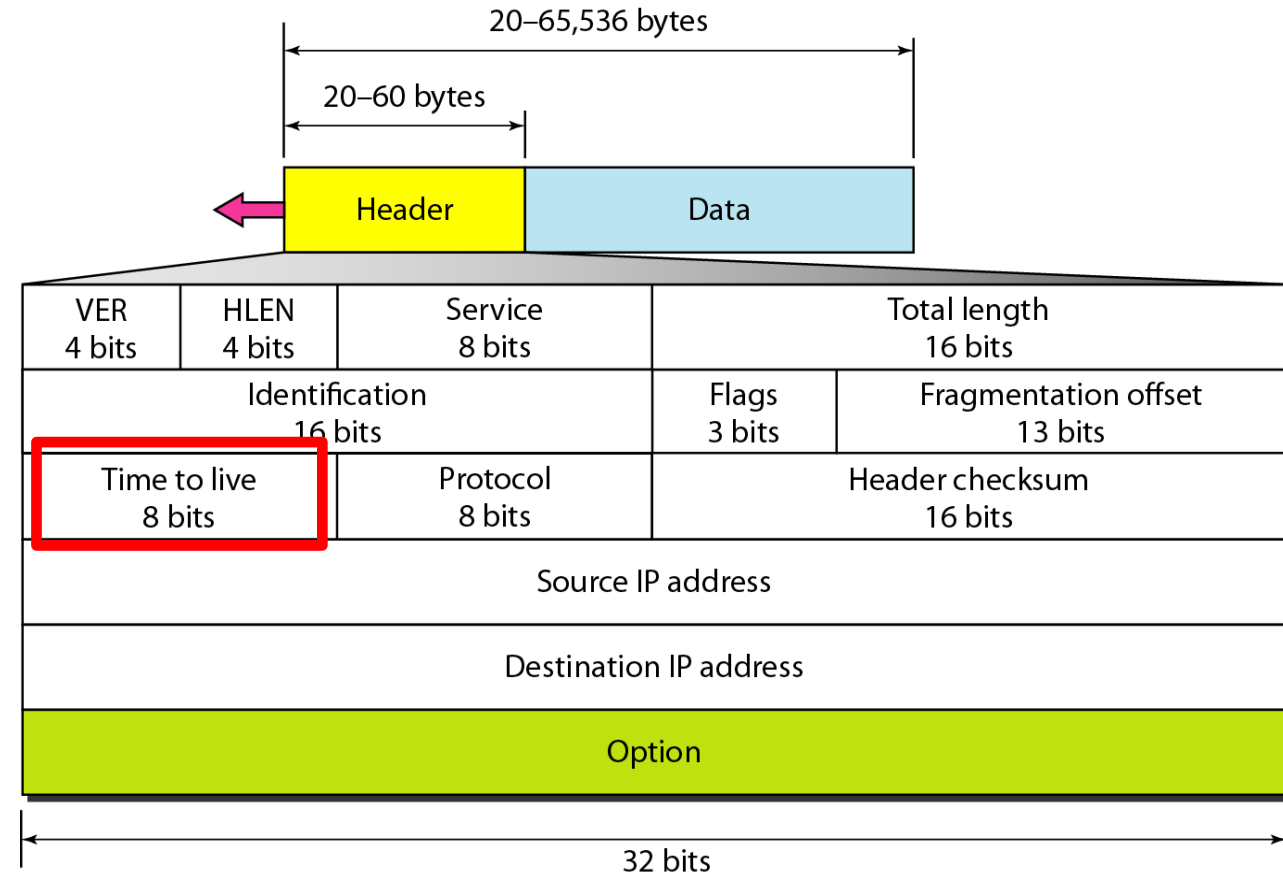
TIME TO LIVE (TTL) FIELD

TTL is an **8-bit** field that limits a datagram's lifetime

Helps **prevent** datagrams from persisting (e.g. going in circles) on an internet

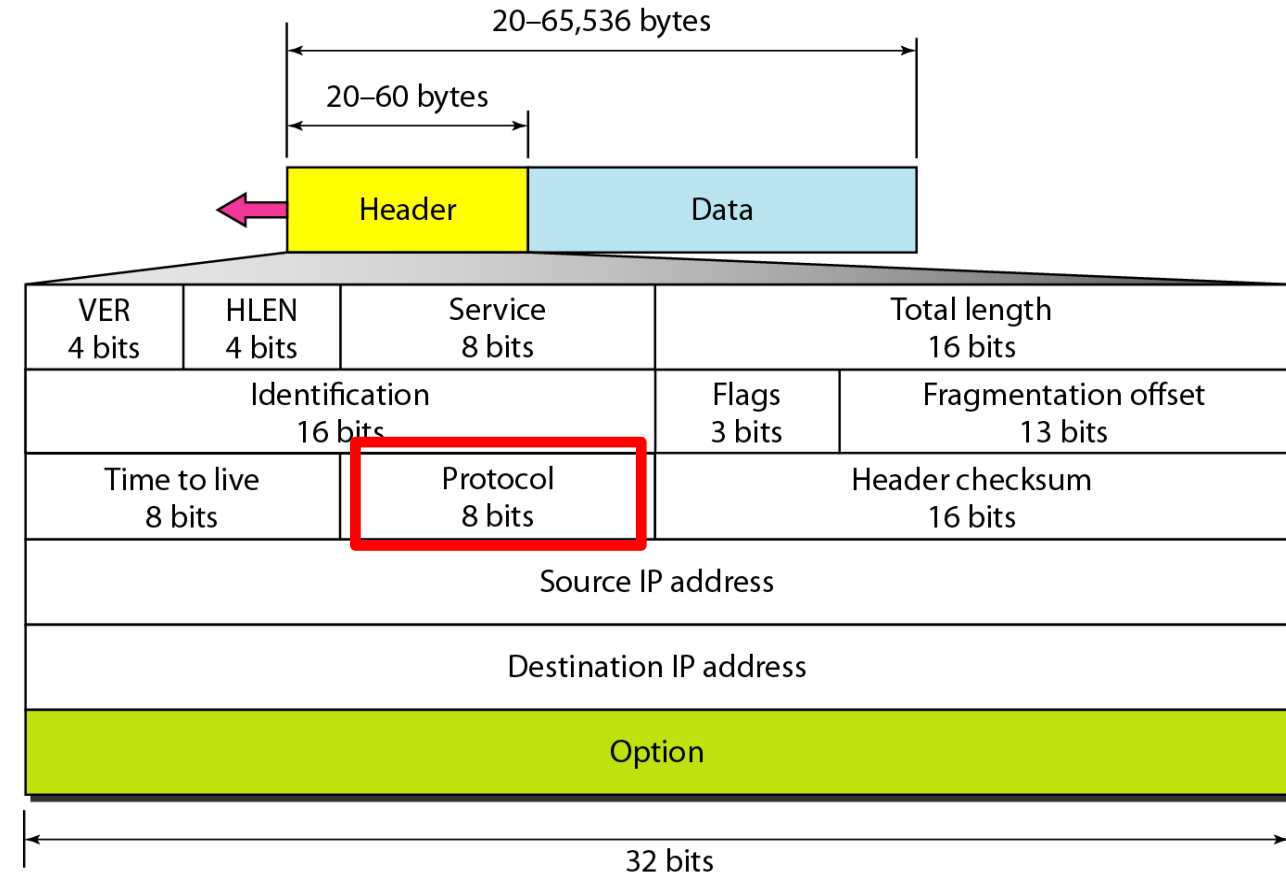
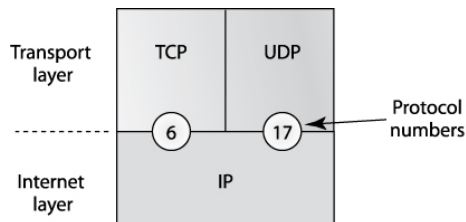
When the datagram arrives at a router, the router **decrements** the TTL field by one

When the TTL field hits **zero**, the router discards the packet and typically sends a ICMP Time Exceeded message to the sender



PROTOCOL FIELD IN IP HEADER

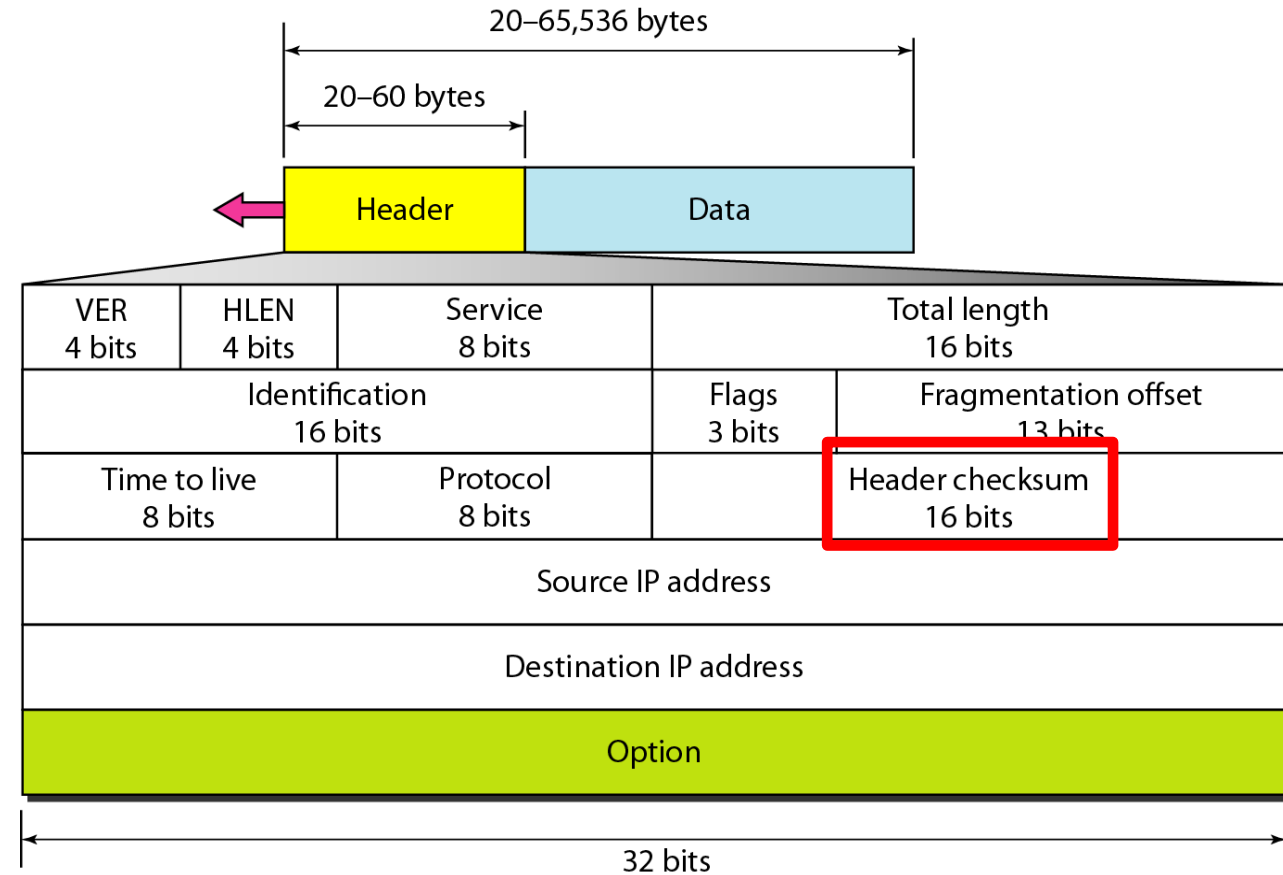
Value	Protocol
1	Internet Control Message Protocol (ICMP)
2	Internet Group Message Protocol (IGMP)
4	IP in IP
6	Transmission Control Protocol (TCP)
17	User Datagram Protocol (UDP)
46	Resource Reservation Protocol (RSVP)
88	Enhanced Interior Gateway Routing Protocol (EIGRP)
89	Open Shortest Path First (OSPF)



HEADER CHECKSUM FIELD

Checksum protects against the corruption that may occur during the transmission of a packet

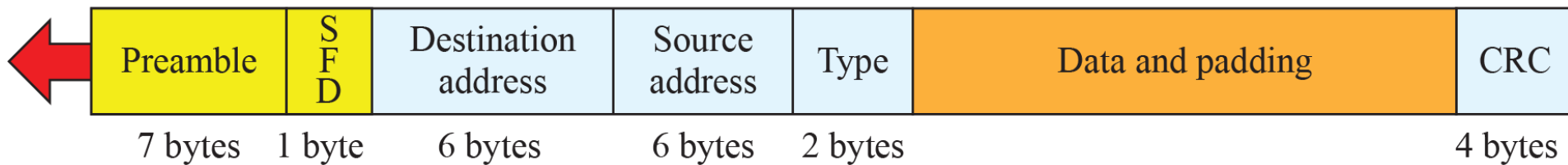
Covers only the header, not the data.



ETHERNET FRAME

Preamble: 56 bits of alternating 1s and 0s

SFD: Start frame delimiter, flag (10101011)



Preamble: A 7-byte sequence that helps receivers synchronize their clocks.

SFD: A 1-byte field that signals the start of the frame.

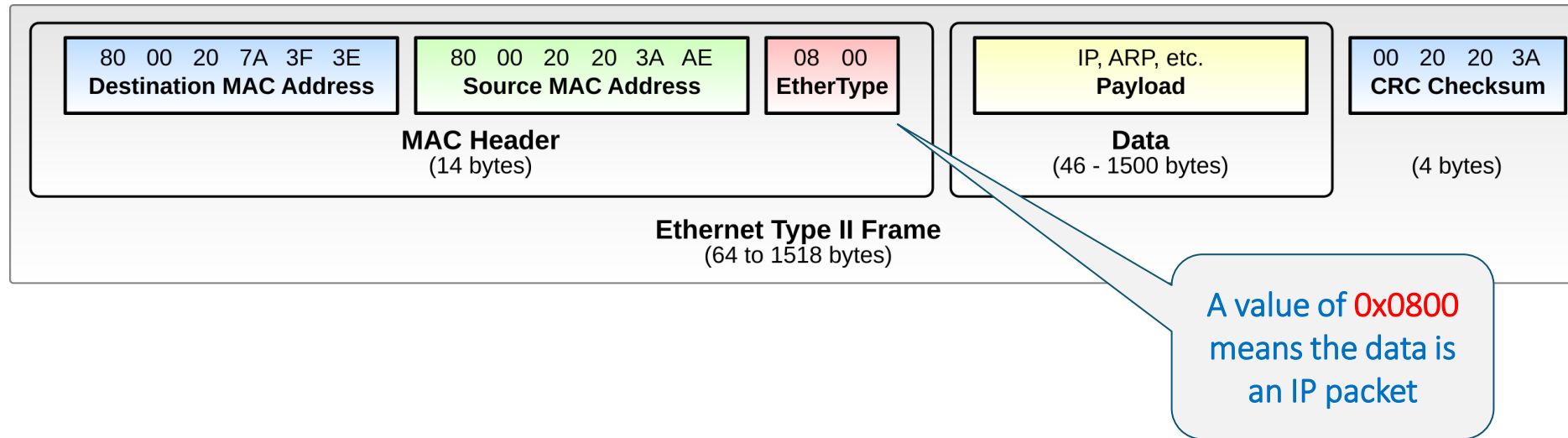
Destination and Source address: A 6-byte field specifying the hardware address.

Type: In Ethernet II frames, this 2-byte field identifies the protocol encapsulated in the payload (e.g., 0x0800 for IPv4).

Data and Padding: Contains the encapsulated data from the higher-layer protocol.

A 4-byte cyclic redundancy check (**CRC**) used to detect errors in the transmitted frame.

IPV4 ENCAPSULATION IN ETHERNET



- + Frame 32: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface 0
- Ethernet II, Src: IntelCor_e4:4f:0c (a0:88:b4:e4:4f:0c), Dst: D-LinkIn_db:03:31 (1c:af:f7:db:03:31)
 - + Destination: D-LinkIn_db:03:31 (1c:af:f7:db:03:31)
 - + Source: IntelCor_e4:4f:0c (a0:88:b4:e4:4f:0c)
Type: IP (0x0800)
- + Internet Protocol Version 4, Src: 172.17.0.106 (172.17.0.106), Dst: 172.17.0.1 (172.17.0.1)
- + User Datagram Protocol, Src Port: 61532 (61532), Dst Port: domain (53)
- + Domain Name System (query)

IPV6 ENCAPSULATION IN ETHERNET

Type value: 0x86DD

- + Frame 90: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface 0
- Ethernet II, Src: D-LinkIn_db:03:31 (1c:af:f7:db:03:31), Dst: IntelCor_e4:4f:0c (a0:88:b4:e4:4f:0c)
 - + Destination: IntelCor_e4:4f:0c (a0:88:b4:e4:4f:0c)
 - + Source: D-LinkIn_db:03:31 (1c:af:f7:db:03:31)
 - + Type: IPv6 (0x86dd)
- + Internet Protocol Version 6, Src: fe80::1eaf:f7ff:fedb:331 (fe80::1eaf:f7ff:fedb:331), Dst: fe80::88fe:f017
- + Internet Control Message Protocol v6

TRANSMISSION CONTROL PROTOCOL (TCP)

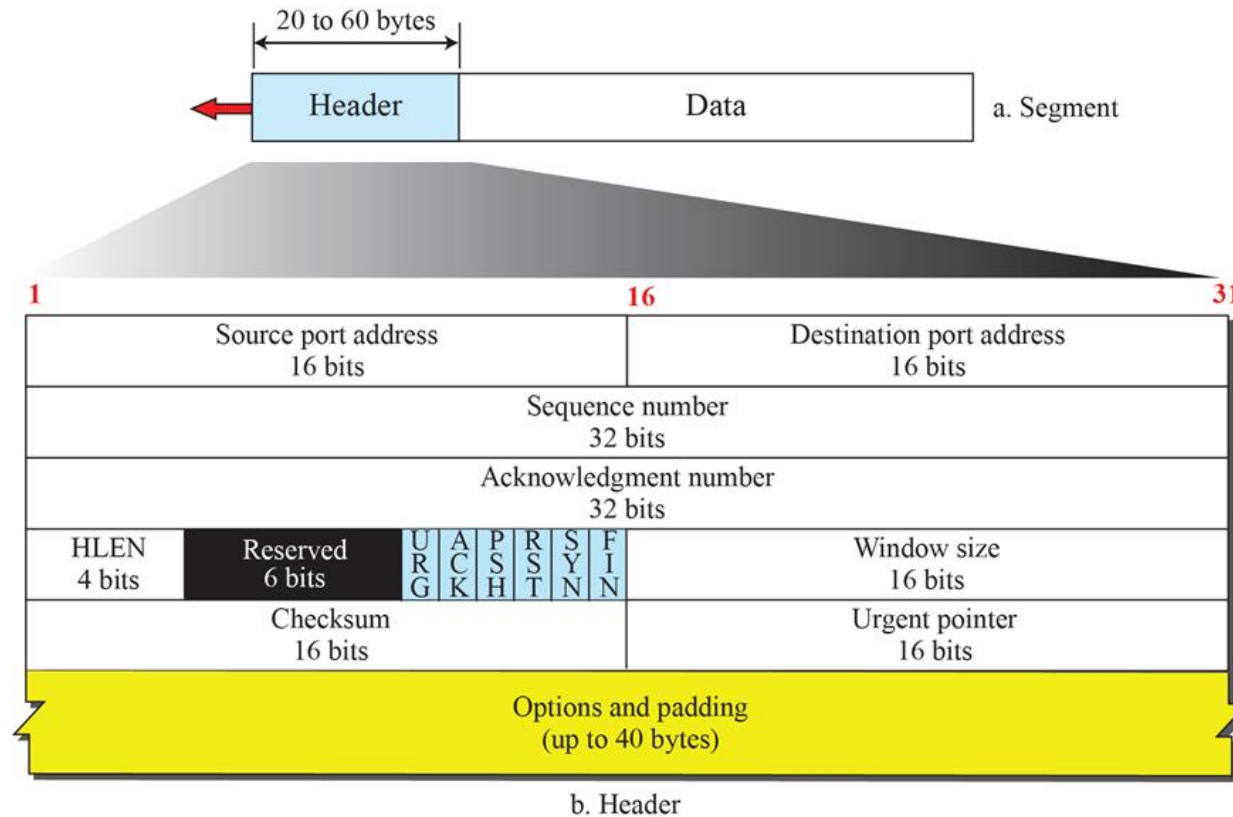
Transmission Control Protocol (TCP) is a **core network protocol** that is used to facilitate communication and data transfer between two or more endpoints.

Transmission Control Protocol (TCP) is a **reliable, connection-oriented** protocol that provides **in-order** delivery of data packets

TCP provides reliability using **error checking, acknowledgement** and **retransmission**

Suitable for **applications** like web browsing, file transfer, email, online banking, VPNs.

TCP HEADER



Sequence Number Indicates the byte number of the first data byte in this segment.

Acknowledgment Number indicates the next expected byte. Confirms receipt of data.

Header Length specifies TCP header size in 32-bit words.

Window Size specifies how much data the receiver can accept.

Checksum used for error detection.

Flag	Name	Purpose
URG	Urgent	Urgent pointer is valid
ACK	Acknowledgment	ACK field is valid
PSH	Push	Deliver data immediately
RST	Reset	Abort/reset connection
SYN	Synchronize	Establish connection
FIN	Finish	Terminate connection

THE TRANSMISSION CONTROL PROTOCOL (TCP)

TCP layer creates connection between sender and receiver using **port numbers**

- The port number **identifies** a particular application on a particular device (IP address)

When two endpoints establish a connection, TCP begins by performing a **three-way handshake** between the two endpoints.

TCP can **multiplex** multiple connections (using port numbers) over a single IP line

THE TRANSMISSION CONTROL PROTOCOL (TCP)

To ensure reliable (in-order) delivery, TCP will utilize a **sequence number** system.

TCP also utilizes a **retransmission** system to ensure that any lost or corrupted data is resent.

- Acknowledgement

TCP also utilizes a **sliding window** mechanism to manage the flow of data (end-to-end flow control)

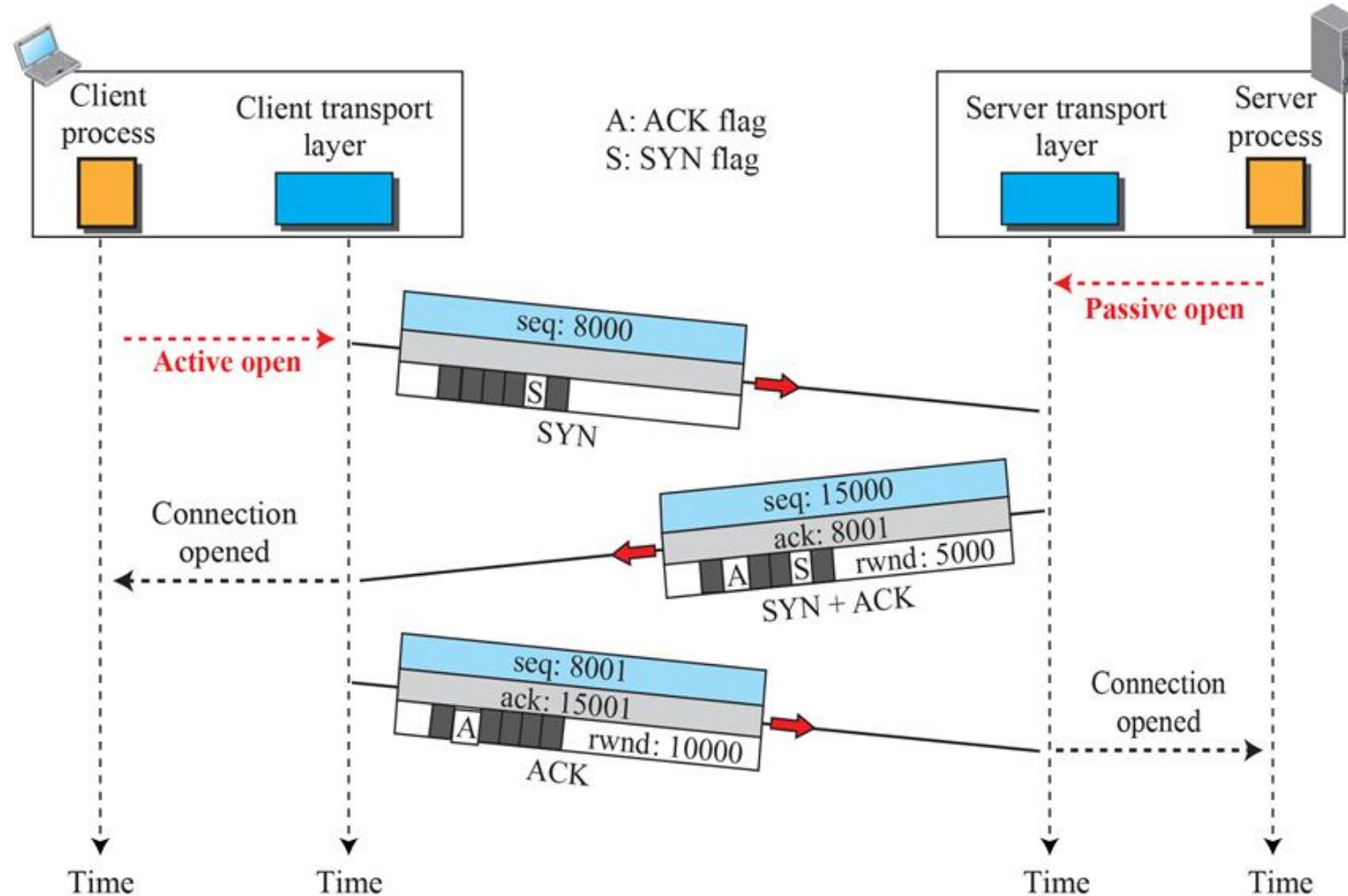
TCP can perform end-to-end **error correction**

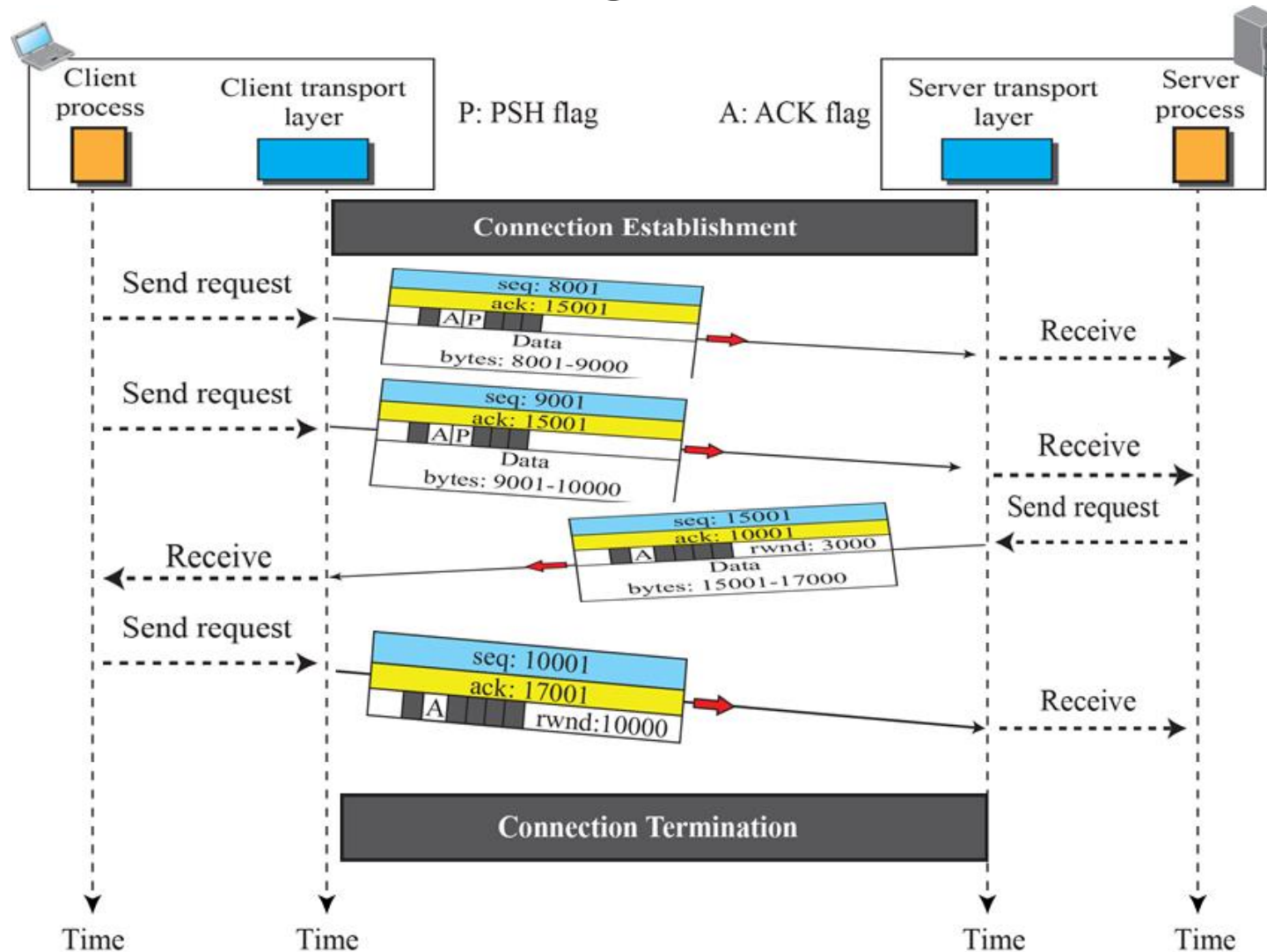
- Checksum

TCP allows for the sending of **high priority data**

- Urgent Pointer

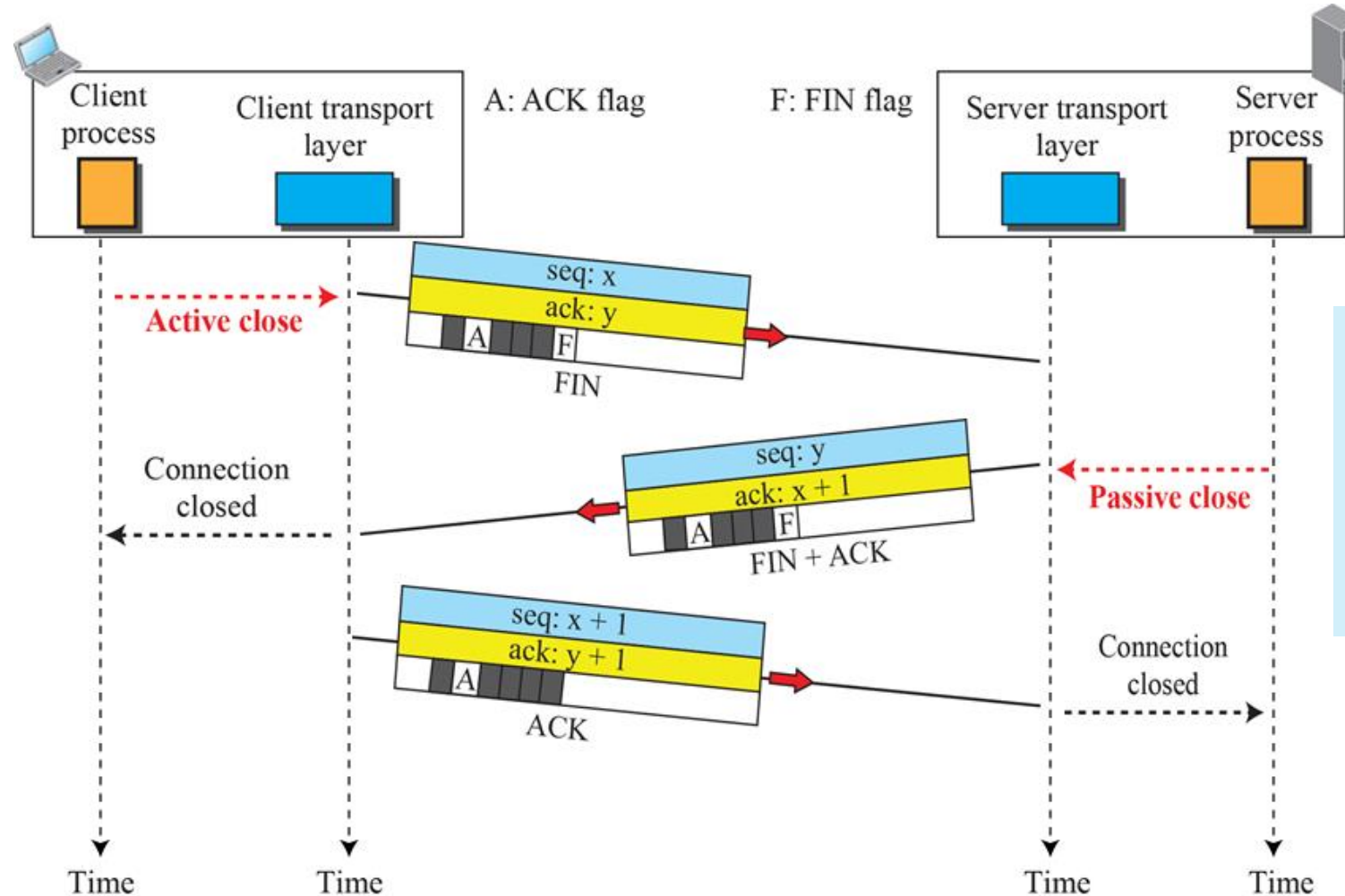
CONNECTION ESTABLISHMENT USING THREE-WAY HANDSHAKING





- **Cumulative Acknowledgment:** TCP acknowledges all bytes up to a certain sequence number, reducing the need for individual ACKs for each segment.
- **Piggybacking:** TCP combines acknowledgments with outgoing data segments to improve efficiency and reduce overhead.

CONNECTION TERMINATION USING THREE-WAY HANDSHAKING



TCP connection termination uses a FIN-ACK exchange in both directions to gracefully close a connection, ensuring all data is delivered before shutdown.

USER DATAGRAM PROTOCOL (UDP)

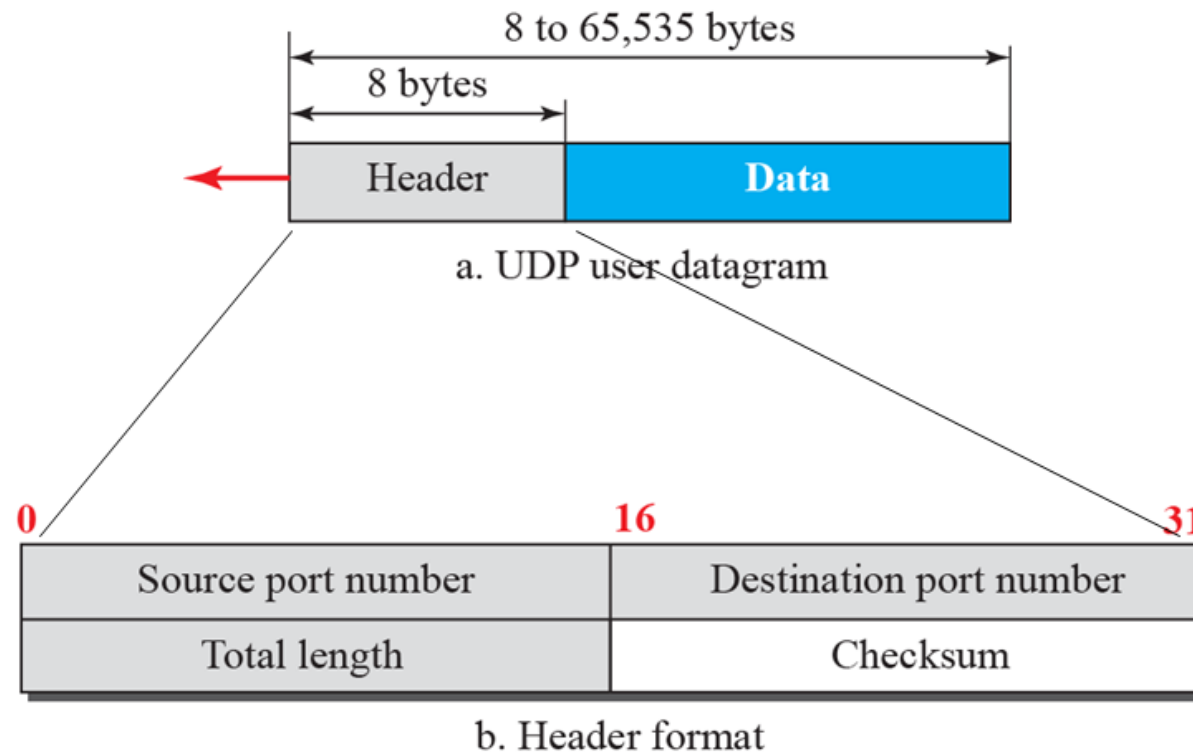
The User Datagram Protocol (UDP) is a **connectionless, unreliable** transport protocol

UDP is a very simple protocol using a minimum of **overhead**

UDP packets, called user datagrams, have

- **Fixed-size header** of 8 bytes made of four fields, each of 2 bytes (16 bits)
 - The first two fields define the **source** and **destination port numbers**
 - The third field defines the total **length** of the user datagram, header plus data

UDP HEADER



UDP APPLICATIONS

Unlike TCP, UDP **does not guarantee** data delivery or ensure that the data arrives in the correct order.

UDP is suited for applications where low **latency** and reduced **overhead** are more important than reliability and ordered data transmission.

- Example: Streaming media, VOIP, Online gaming, etc.

NETWORK TRAFFIC ANALYSIS



ACTIVITY: OBSERVING THE TCP THREE-WAY HANDSHAKE AND DATA TRANSMISSION

Objective: Observe how a TCP connection is established, used for data transmission, and terminated.

To-do:

- Start a TCP listener using netcat.
- Connect to the listener using a netcat client.
- Capture local TCP traffic using tcpdump.
- Identify the TCP three-way handshake.
- Observe TCP data transmission.
- Recognize basic TCP flags such as SYN, SYN-ACK, ACK, PSH, FIN, and RST.
- Explain the role of the loopback interface in local network communication.

NETWORK VISIBILITY

Security analysts rely heavily on **network visibility**, the ability to see, trace, and analyze all traffic across digital environments, to accurately evaluate and improve an organizational **security posture**.

Why this relationship is **crucial**?

- Eliminating blind spots
- Detecting lateral movement
- Real-time threat analysis
- Effective incident response

PACKET CAPTURE

Packet capture is the process of collecting and recording network packets as they pass through a network interface or monitoring point.

Packet capture is useful when analysts need detailed **evidence**.

Packet capture can be performed using **tools** such as:

- Tcpdump
- Wireshark
- Tshark
- Suricata
- Zeek
- Security Onion

PACKET CAPTURE

To **intercept** packets, defenders must place the monitoring tool or sensor where traffic can be observed.

- **Port Mirroring / SPAN Ports** – a switch feature that copies traffic to a monitoring port.
- An **inline network device** sits directly in the path of network traffic. Traffic must pass through the device before reaching its destination. E.g. Firewall, IPS etc.
- A **network tap** is a dedicated hardware device used to copy network traffic for monitoring. Unlike SPAN ports, taps are purpose-built for reliable traffic capture.

PCAP

A **PCAP** (Packet Capture) file is a data file used to store network traffic captured during packet sniffing.

- **Records** the raw data packets traveling across a network, allowing security teams to analyze network behavior, troubleshoot issues, and detect security threats.
- **Preserve** everything from packet payloads to exact timestamps.
- **Resource** intensive and hard to **scale**.

```
# Captures traffic from interface eth0 and saves it to a file.  
sudo tcpdump -i eth0 -w traffic.pcap  
# Reads pcap file  
tcpdump -r traffic.pcap  
wireshark traffic.pcap
```

FLOW RECORD

A **flow record** is a summarized record of a network conversation.

- **Common fields** include, source and destination IP addresses, ports, protocols, timestamps, and total byte/packet counts

Generated by network devices or monitoring sensors.

Efficient bandwidth and storage **requirements**

High-level pattern and anomaly detection

Flow Record Example:

Start time:	10:15:22
Duration:	3.8 seconds
Source IP:	192.168.1.10
Source port:	49152
Destination IP:	203.0.113.50
Destination port:	443
Protocol:	TCP
Packets:	18
Bytes:	12,420
State:	Established

NETWORK TRAFFIC ANALYSIS

```
# Find interfaces on the machine.
```

```
$ ip link show
```

```
# Capture and display all packets passing through the loopback interface
```

```
$ sudo tcpdump -i lo -X
```

```
# Capture traffic involving example.com:
```

```
$ sudo tcpdump -i enp0s3 -n host example.com
```

```
# Capture Traffic Going to a Specific Destination
```

```
$ sudo tcpdump -i enp0s3 -n dest <IPofGoogle.com>
```

```
# Save traffic to PCAP file.
```

```
$ sudo tcpdump -i enp0s3 -n -w ~/capture.pcap
```

Task to To-Do

You're working as a SOC analyst.

Capture only traffic from your machine to either **DNS port 53** or **HTTPS port 443**, save it to a PCAP file, and then read back only the HTTPS traffic from that saved file.

NETWORK TRAFFIC ANALYSIS

PCAP Analysis of a Suspected Malware Download

You are working as a SOC analyst and have been given a packet capture file named `tcpdump-Sample01.pcap`.

This PCAP contains network traffic from a suspected compromised endpoint (10.0.0.168).

Your task is to investigate (using **tcpdump**) the capture, identify suspicious HTTP activity, determine whether a malicious file was downloaded, and document the evidence supporting your conclusion.

NETWORK TRAFFIC ANALYSIS

PCAP Analysis of a Suspected Malicious Macro-Enabled Word Document Infection

You are working as a SOC analyst. You have been given a packet capture file named: `tcpdump-Sample02.pcap`

This PCAP contains network traffic from a suspected compromised endpoint. Initial reporting suggests that the victim may have opened a **malicious macro-enabled Microsoft Word document**, which resulted in suspicious outbound communication.

Your task is to investigate the PCAP, identify the affected internal host, analyze its communication with suspicious external infrastructure, extract relevant indicators of compromise, and determine whether the observed activity is benign, suspicious, or malicious.

NETWORK TRAFFIC ANALYSIS

Wireshark is a graphical network packet analyzer.

- Captures packets from a network interface or opens saved packet capture files such as .pcap and .pcapng, then decodes the traffic so analysts can inspect protocols, headers, fields, payloads, timing, and conversations.

Used extensively for Incident Triage, Malware Analysis, Forensic Evidence and for Protocol Validation

Filters: Capture filters and display filters in Wireshark control when and where you filter network data.

- Capture filters determine which packets are saved to your computer's memory or disk during a live capture.
- Display filters change what you see on your screen from a file that has already been captured (or is actively being captured).

NETWORK TRAFFIC ANALYSIS

Wireshark Statistics refers to built-in analysis views that summarize the packet capture.

- Instead of manually inspecting every packet, analysts can use the Statistics menu to quickly understand the capture at a higher level.

Commonly used statistics:

- Capture file properties
- Protocol hierarchy
- Conversations
- Endpoints
- I/O graphs
- Flow graph

NETWORK TRAFFIC ANALYSIS

PCAP Analysis of a Suspicious executable download

You are working as a SOC analyst. You have been given a packet capture file named:wireshark-Sample01.pcap

The capture contains network traffic from a suspected compromised Windows endpoint. Your task is to investigate the PCAP in Wireshark, identify suspicious HTTP activity, determine whether a malicious executable was downloaded, and document the evidence that supports your conclusion.

NETWORK TRAFFIC ANALYSIS

Answer the following questions:

- What is the total number of packets in the capture?
- Which internal IP address appears to be the main endpoint in this capture?
- Which external IP addresses communicate with this internal host?
- Which conversation appears most relevant for further investigation?
- View protocol hierarchy, which major protocols are present?
- Which external servers received HTTP requests from internal host?
- Do any request paths look unusual?
- Do any requests appear to involve downloading a file?
- Are any requested files executable or suspicious?
- Is there any executable file requested over HTTP? What is the filename, request path, external IP address hosted the file, full URL?
- Was the file downloaded from a trusted or expected domain?

NETWORK TRAFFIC ANALYSIS

Answer the following questions:

- Follow TCP stream and note:
 - HTTP method, Host, requested file path, User-Agent, response from the server.
 - Does the stream indicate that the executable was delivered to the client?
- Justify why the file .audiodg.exe is suspicious?
- Are there HTTP POST requests after the suspicious download? What User-Agent is associated with the POST traffic?
- Why might HTTP POST activity after a suspicious executable download be concerning?
- Validate your findings using reputation sources such as VirusTotal, WHOIS, etc.

NETWORK TRAFFIC ANALYSIS

PCAP Analysis of Unknown Suspicious Network Activity

You are working as a SOC analyst. You have been given a packet capture file: `wireshark-Sample02.pcap`

No specific indicator of compromise has been provided. Your task is to use Wireshark to investigate the PCAP, identify the suspicious activity, determine the likely victim host, and explain whether the traffic indicates malware infection or normal network activity.

You must begin with broad PCAP review and narrow the investigation based on evidence discovered in Wireshark.

References

- <https://www.malware-traffic-analysis.net/>
- <https://unit42.paloaltonetworks.com/feb-wireshark-quiz-answers/>
- <https://attack.mitre.org/software/S0650/>

NETWORK TRAFFIC ANALYSIS (WIRESHARK CHALLENGE)

Your organization received an alert indicating unusual traffic patterns and a large number of repeated queries originating from one of its external-facing endpoints (172.16.1.16). Analyze the **wireshark_challenge.pcap** packet capture and answer the following questions:

1. How many HTTP packets are contained in the capture file?
2. What relative path victim accessed on the web server to request a file download?
3. Based on the response header, what file type or format the downloaded file is?
4. What is the actual file signature (magic bytes) contained in the downloaded file?
5. What command-line utility or program did the victim use to download the file?

NETWORK TRAFFIC ANALYSIS (WIRESHARK CHALLENGE)

Your organization received an alert indicating unusual traffic patterns and a large number of repeated queries originating from one of its external-facing endpoints (172.16.1.16). Analyze the **wireshark_challenge.pcap** packet capture and answer the following questions:

6. What is the SHA-256 hash of the downloaded file?
7. Submit the uncovered hash to VirusTotal. Based on the popular threat label and tags, what type of malware infected the endpoint?
8. Review the domain names queried within the capture. In defanged format, what is the base domain name that is repeatedly queried?
9. Read about MITRE ATT&CK ID T1071.004. What is the commonly used name for the attack technique likely observed in this PCAP file?

REFERENCES

- [1] Murdoch, D. (2024). Blue team handbook: SOC, SIEM, and threat hunting: Practical techniques for security operations and threat hunting teams. O'Reilly Media.
- [2] <https://www.malware-traffic-analysis.net/>
- [3] <https://unit42.paloaltonetworks.com/feb-wireshark-quiz-answers/>
- [4] <https://attack.mitre.org/software/S0650/>

Questions?

